

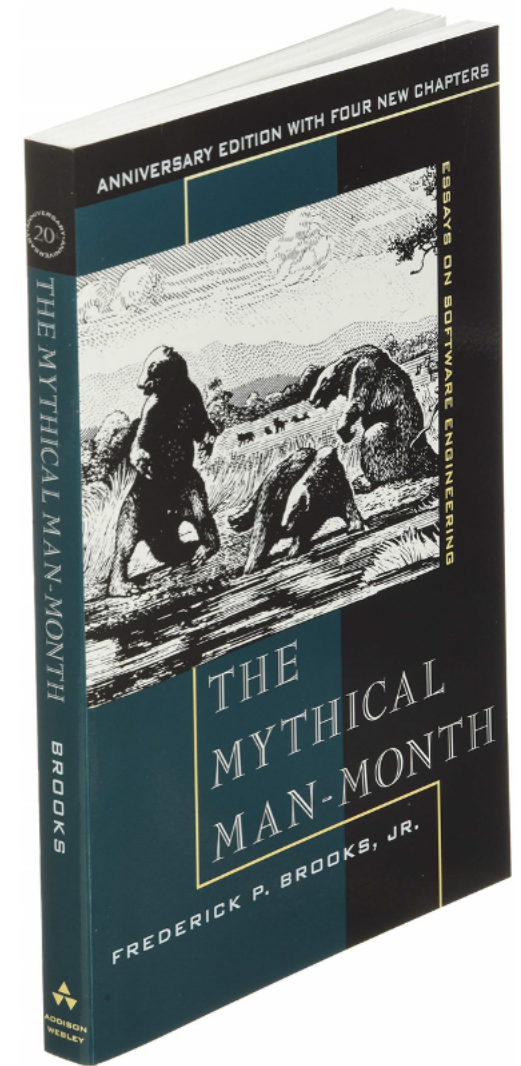
Goals and Requirements

Software Engineering for Intelligent Systems

JOÃO EDUARDO MONTANDON

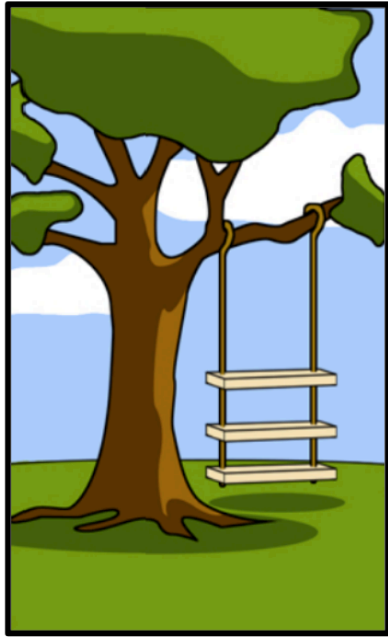
*The hardest single part of building a software system is **deciding precisely what to build.***

— Fred Brooks

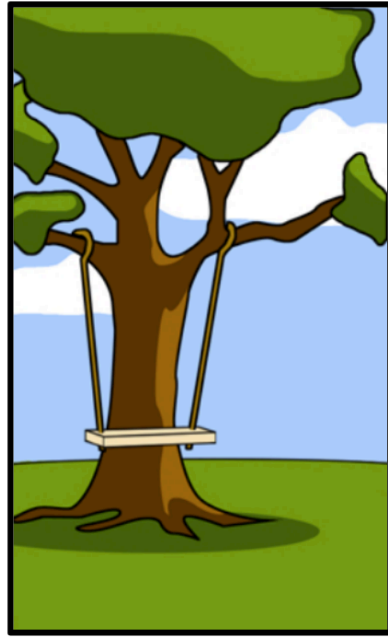


Software Requirements

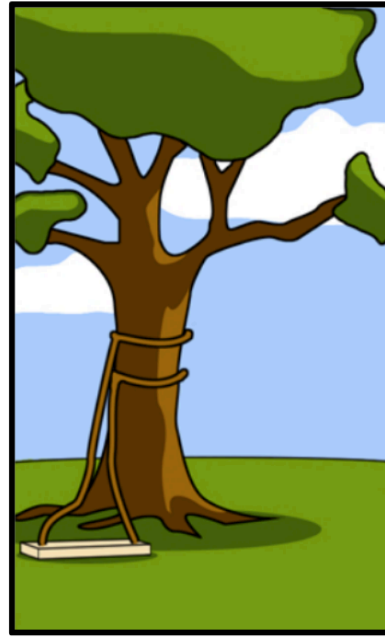
Are we building the right system?



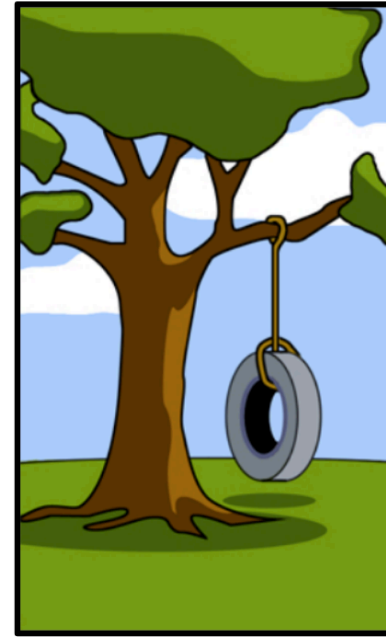
How the customer explained it



How the project leader understood it



How the programmer wrote it



What the customer really needed

Requirements in Software Systems



```
def compute_deductions(agi, expenses):  
    """  
    Compute deductions based on provided adjusted gross income  
    and expenses in customer data.  
  
    See tax code 26 U.S. Code A.1.B, PART VI.  
    Adjusted gross income must be a positive value.  
    Returns computed deduction value.  
    """
```


Requirements in Intelligent Systems

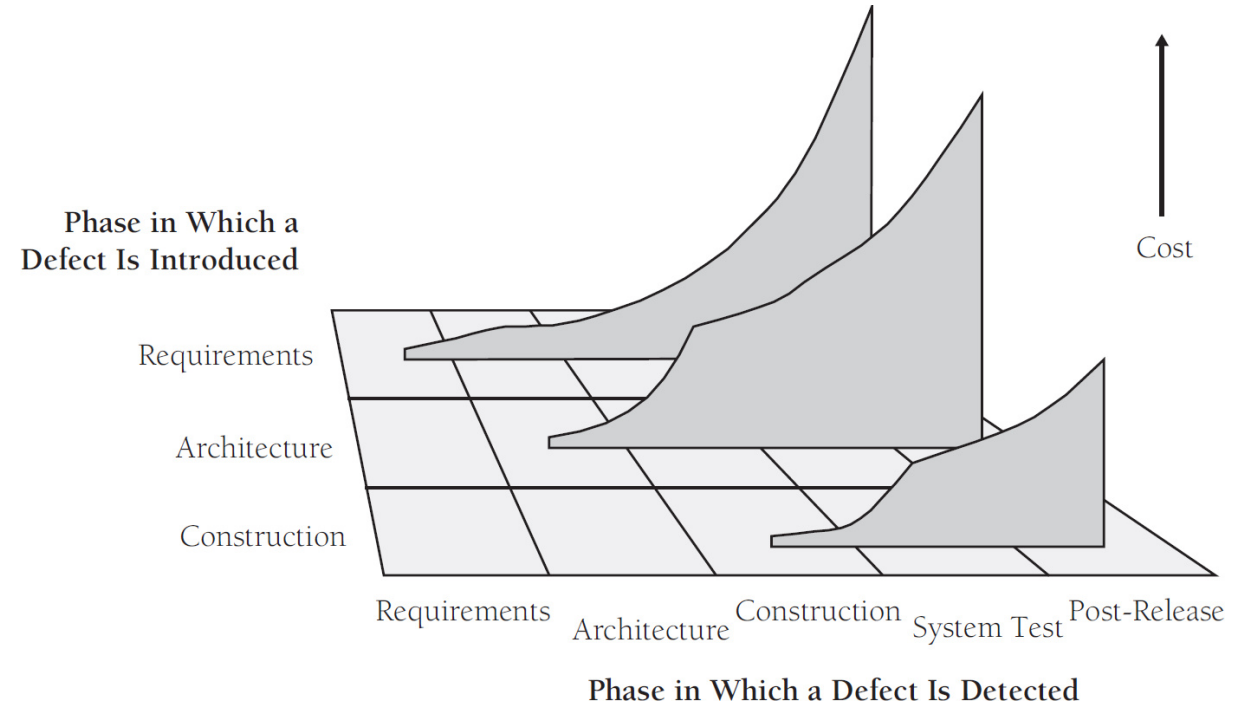


```
def transcribe(audio_file):  
    """  
    Return the text spoken within the audio file.  
  
    ????  
    """
```

The cost of misspecification

The wrong system is the worst mistake we can ever make.

Think before coding. **How?**



Setting Goals

"The first prerequisite you need to fulfill [...] is a clear statement of the problem."

— Steve McConnell, Code Complete

Goals

- Goals are prescriptive statements about intent.
- They provide general information about the purpose of the system according to each stakeholder.
- Should anticipate requirements.
- Can be layered into **different scopes/levels**.

Organization Goals

General goals of the organization.

1. Business: Revenue increase, profit.
 2. Nonprofit: Increasing animal welfare, CO2 reduction, diseases cured, etc.
- Often high-level and long-term, thus difficult to measure.
 - **ML models are not the ultimate goal!**

Product Goals

Concrete outcomes the product should provide.

- Describe what the **product tries to achieve** in terms of behavior and quality.
- First step towards requirements.

Examples:

1. Recommend best route.
2. Transcribe audio files.
3. Summarize text in different languages.

User Goals

How the software is being useful to its users.

- Each user has his own needs, his own goals.
- Should include goals of indirectly affected users.

Example (short-term rental):

- **Guests** want the best price for rental.
- **Hosts** want to maximize the revenue of their houses.
- **Indirect**: local community, municipalities, regulators, etc.

Interesting... 🤔

Model Goals

Quality of the model from the model's perspective.

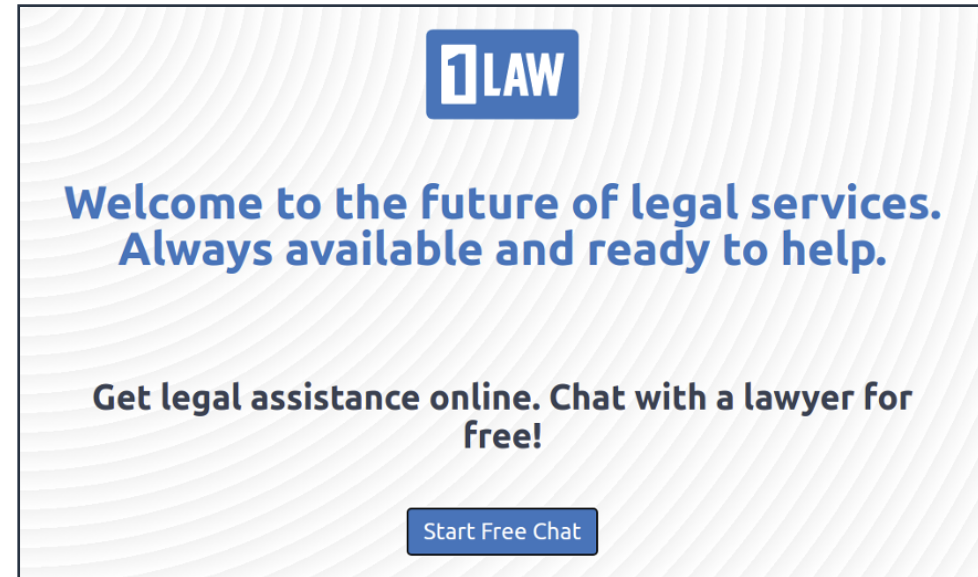
Examples:

- Model accuracy.
- Inference time.
- User engagement.
- Training cost.

Loosely linked with the other goals.

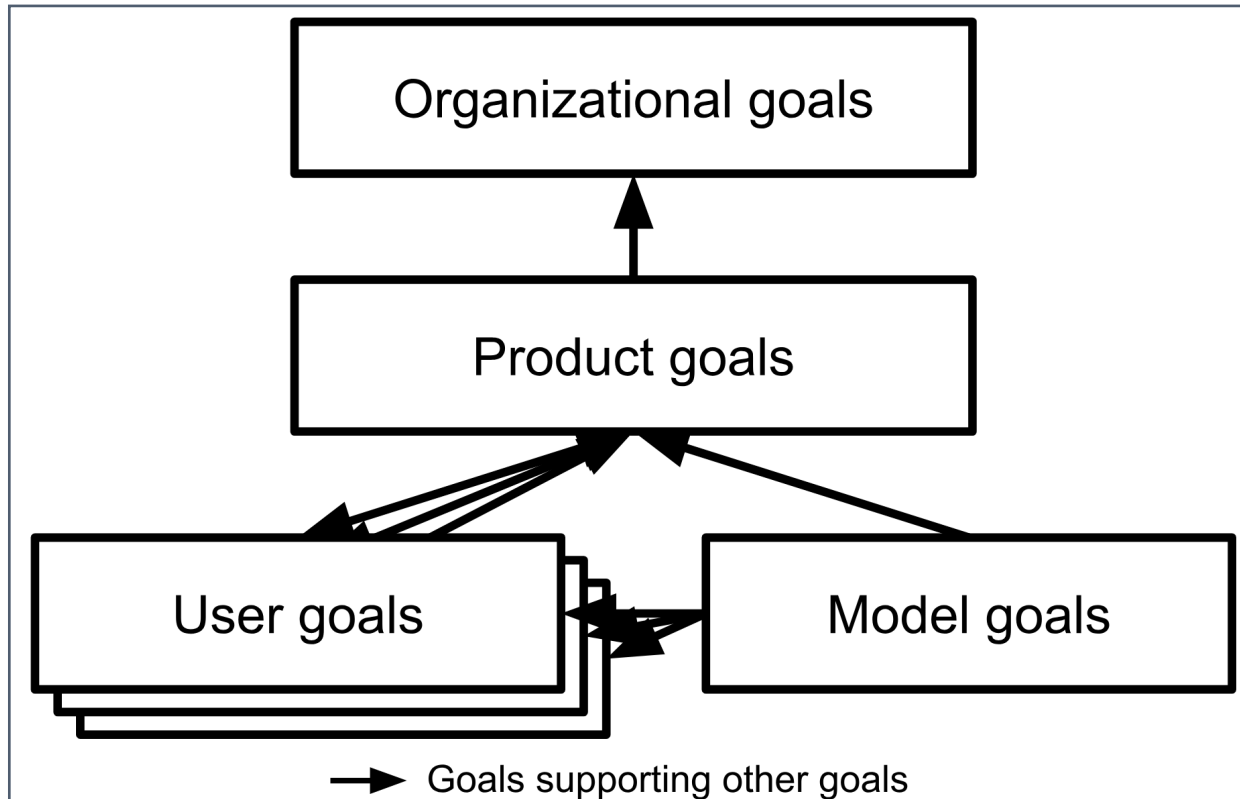
Example: Self-Help Legal Bot

- Your company is offering marketing services to law firms.
- You released a chatbot to attract clients to the firm websites.
- Chatbot provide some guidance to initial clients requests.
- Further info is forwarded to the attorney available.



Organizational goals? System goals? User goals? Model goals?

Goals Relationship

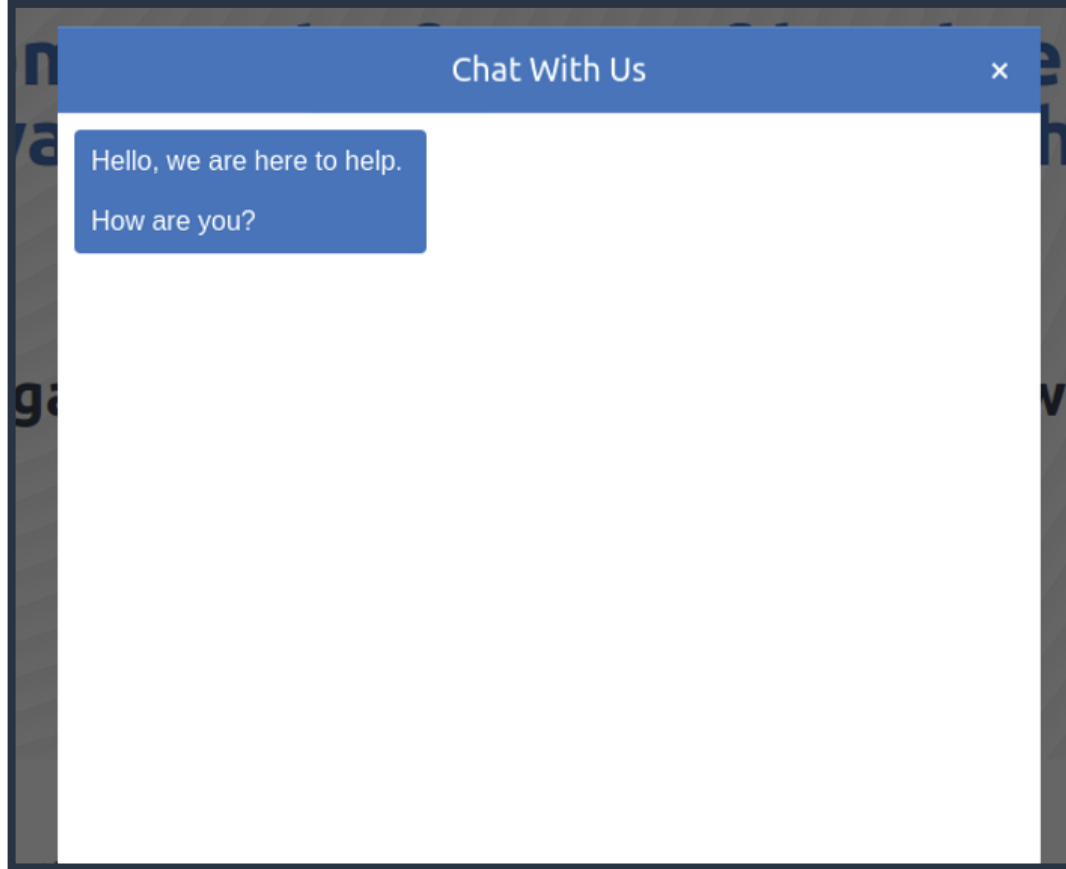


Goals are linked to each other.
Everything is **fine when goals align**.

Better product $\Rightarrow$ happier users $\Rightarrow$ successful organization.

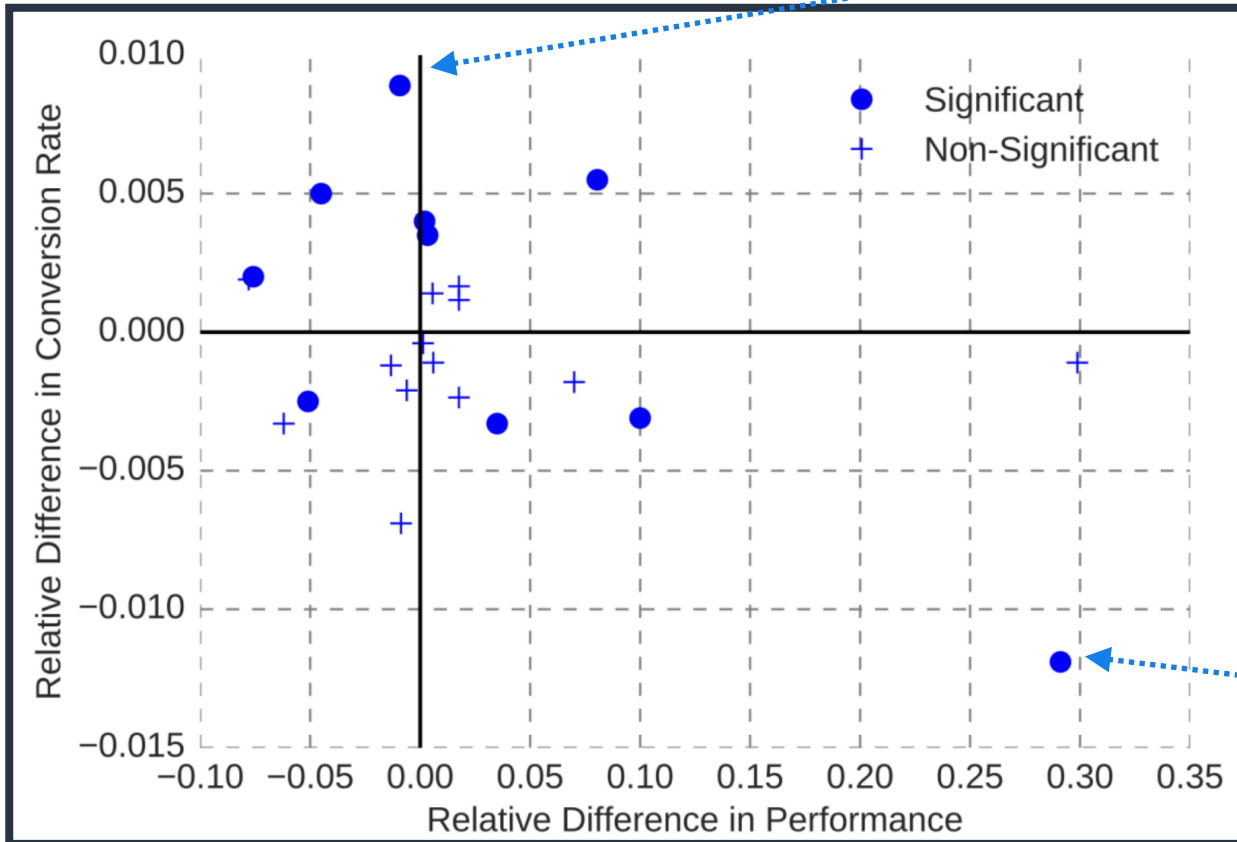
But **goals might not align...**

Conflicting Goals?



The booking.com Case Study

Best Outcome



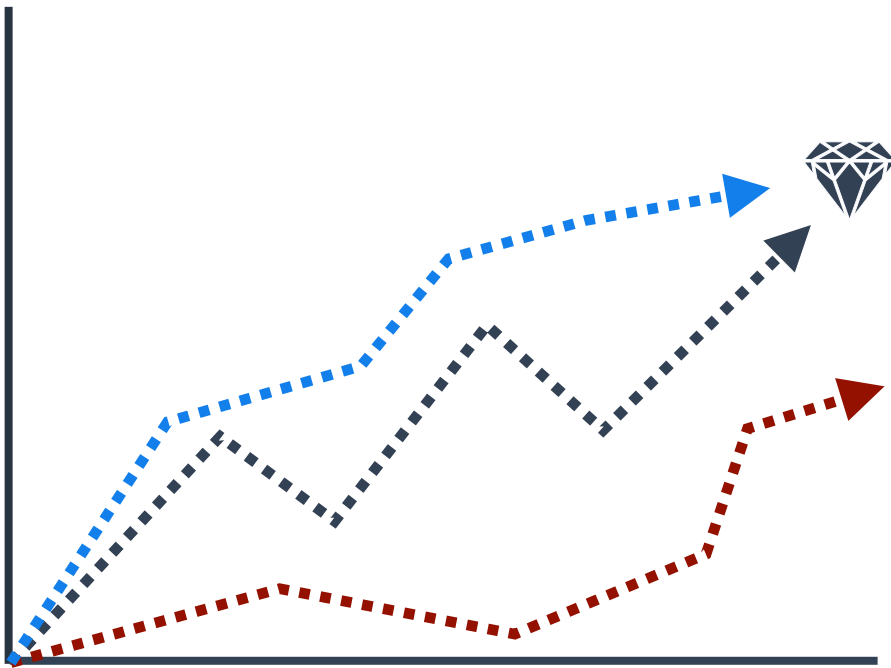
Model accuracy improvement **does not** necessarily **translate to** improvements in **system outcomes**.

"such finding led us to investigate other areas of Booking.com and consistently found the same pattern."

Best Model

Bernardi et al. "150 Successful Machine Learning Models: 6 Lessons Learned at Booking.com."
In Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, 2019

Goals are only useful if measured



If we **care** about a property, then it must be **detectable**.

This way we can distinguish **better** from **worse**.

Quantification: the process of turning observations into numbers.

Defining Measures

Measure: a precise description of the measure to be captured.

Data Collection: a description of what data is collected and how.

Operationalization: A mechanism for computing the measure from the data.

Example: User goal

Measure: Customer satisfaction of subscribing attorneys.

Data Collection: Each month, email 5% of attorneys with a link to an online survey where they can select 1—5 stars rating and provide feedback.

Operationalization: Average the star ratings received from the survey in each month.

Example: Component goal

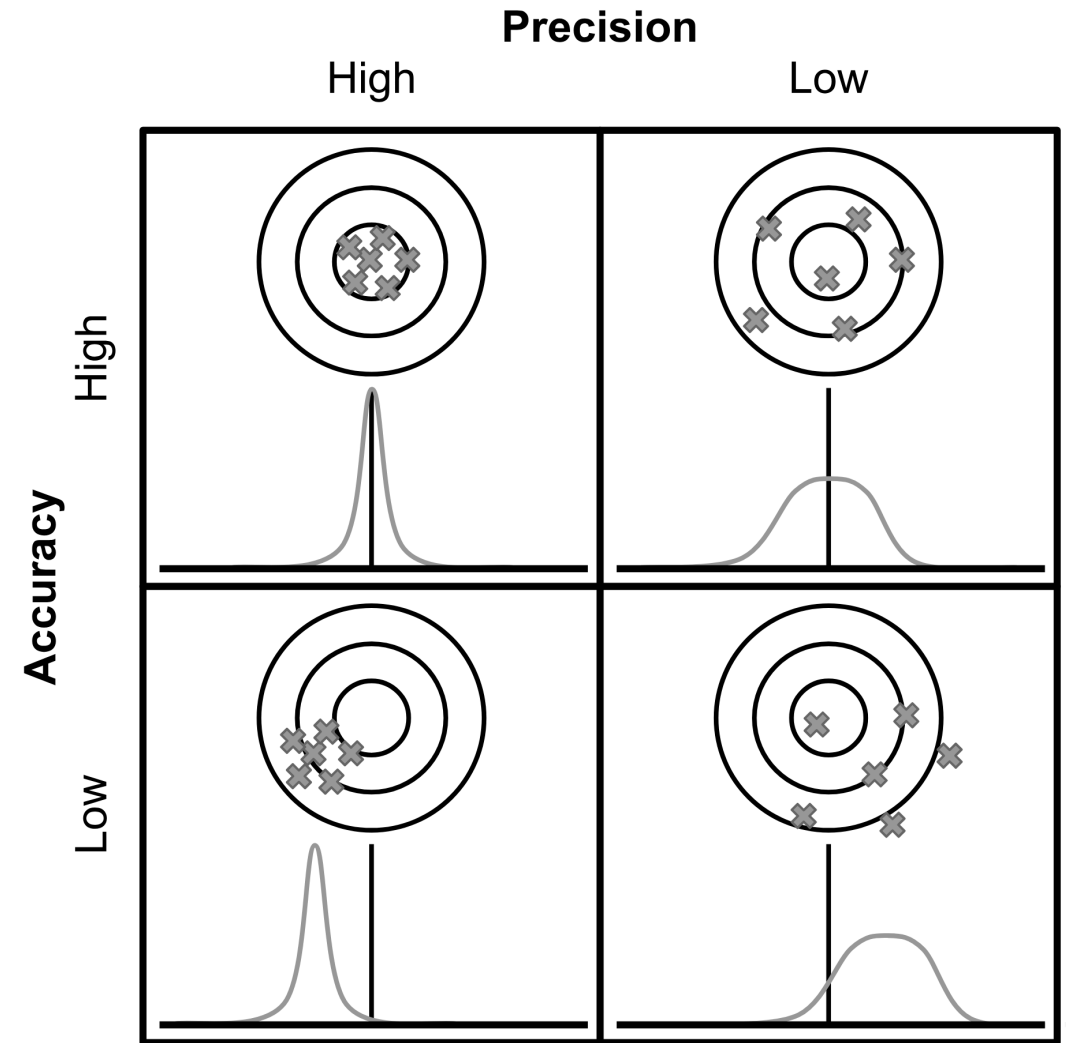
Measure: 90%-quantile response latency of the chatbot.

Data Collection: Log the time of each request and each response in a log file on the server.

Operationalization: Obtain the response latency of each request as $\Delta = \text{resp}_{time} - \text{req}_{time}$. Select the fastest 90% of the requests in the last 24h.

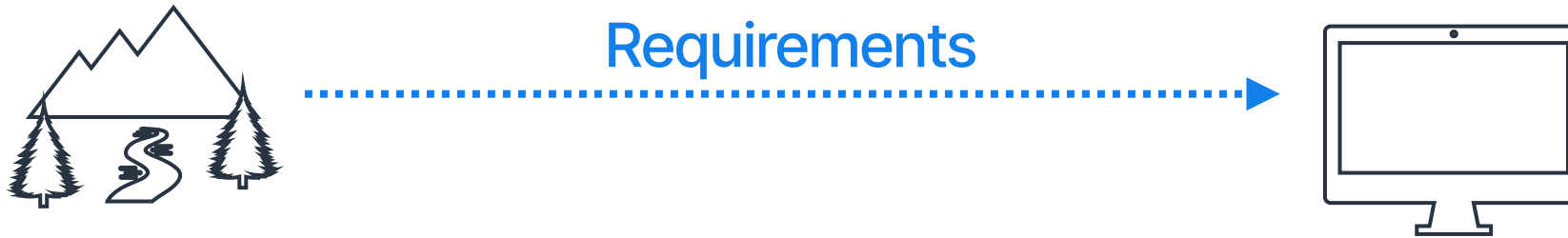
A Good Measure has...

1. **Precision**: measures are not dispersed.
2. **Accuracy**: how close they are to the intended target.
3. **Validity**:
 - A. **Construct**: Are we measuring the intended feature?
 - B. **Predictive**: Can the measure [partially] explain its results?
 - C. **External**: Does the measure generalize beyond initial observations?



Leveraging Requirements

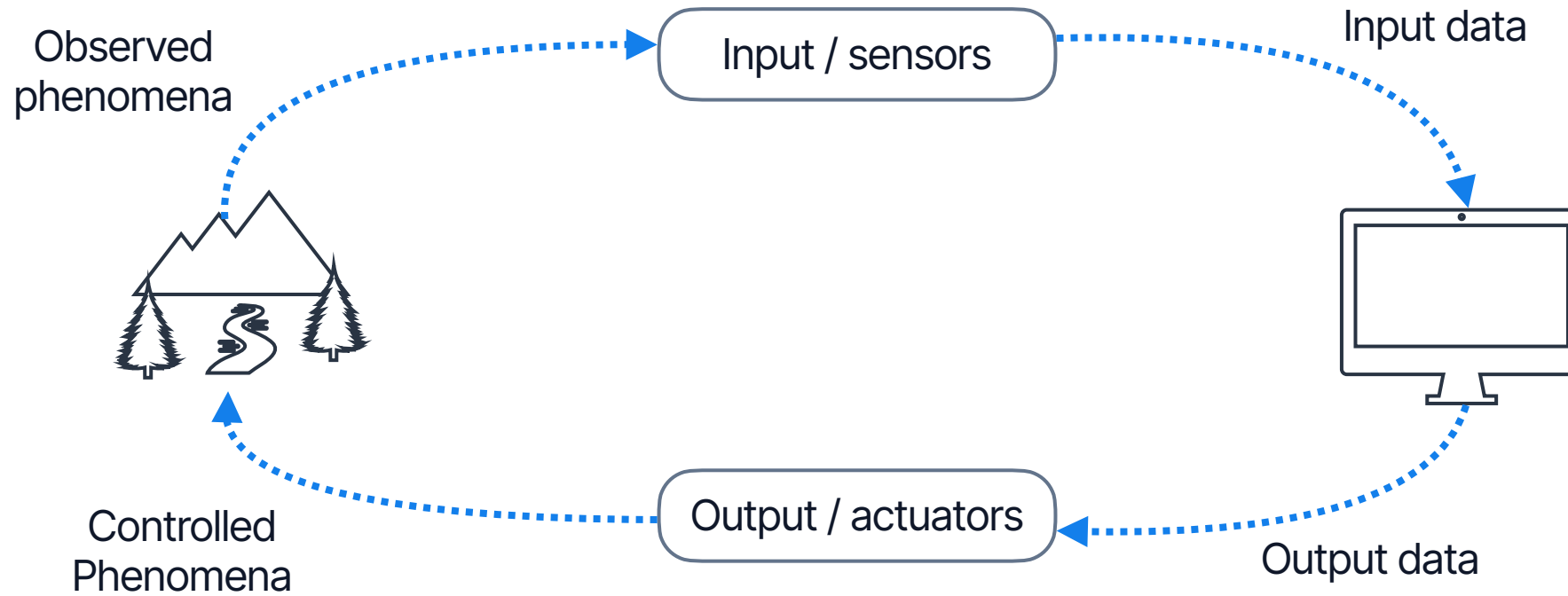
What we want to achieve in the real world
with the software...



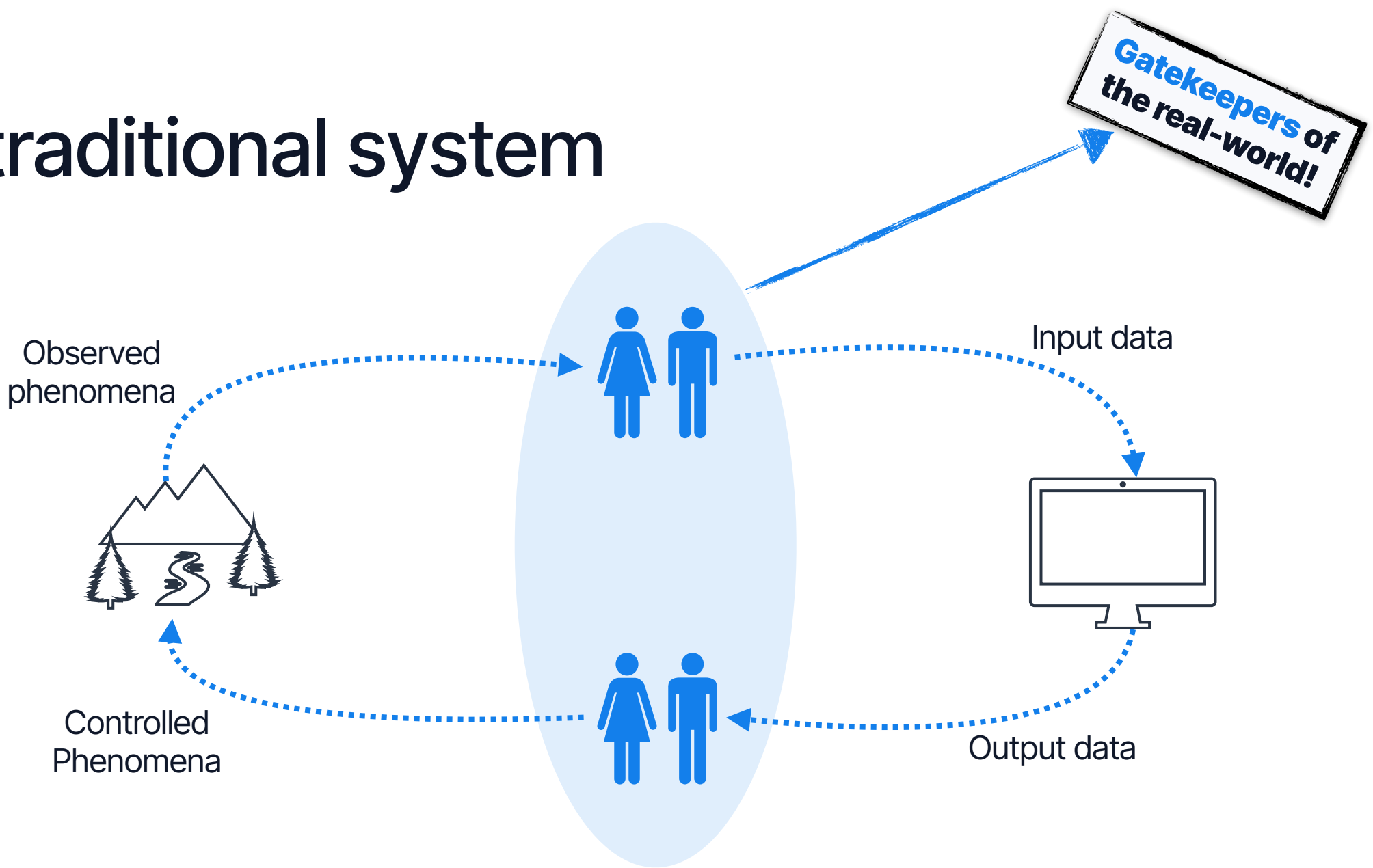
The World and the Machine

- **Software can only interact** with the real world mediated through **input and output devices**.
- **Software cannot reason** directly **about the world**, but only about input data that may be derived from observations about the world made by sensors.
- **Software cannot** directly **influence the world**, but only indirectly by presenting output data to humans or controlling phenomena in the world through actuators.

A closer look



A traditional system

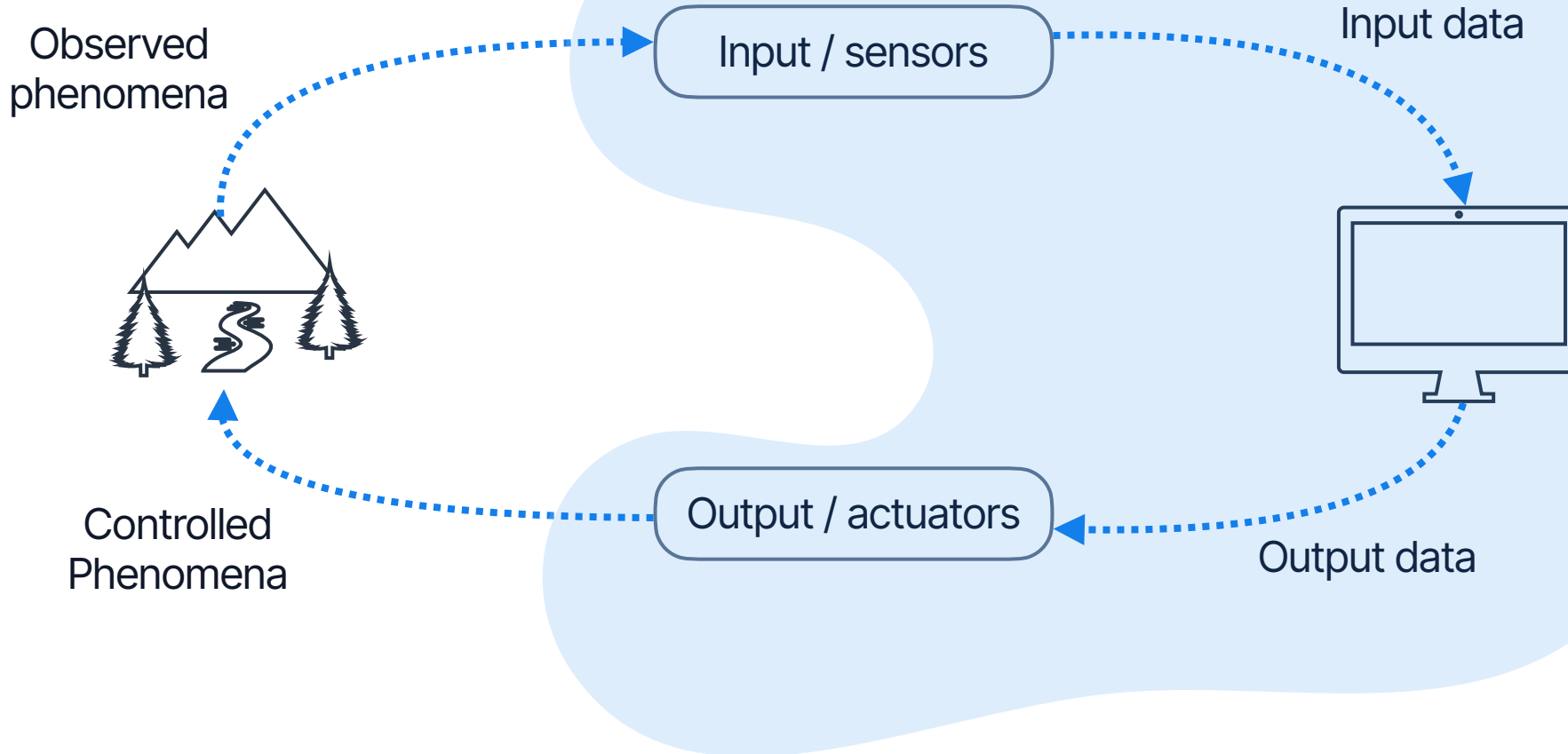


A traditional system

- **Users make sense of the world**, and manages the interaction between software system and outside world.
- **Users decide** which feature to use for a given scenario, and which data to inject in the system.
- **Users interpret** the results and take decisions based on them.

Managing input/output is not system's responsibility!

An intelligent system

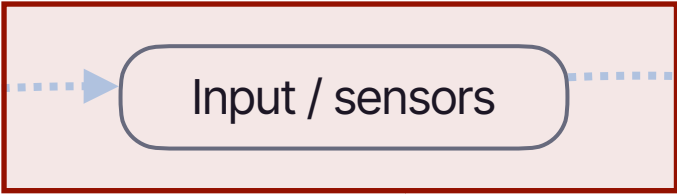
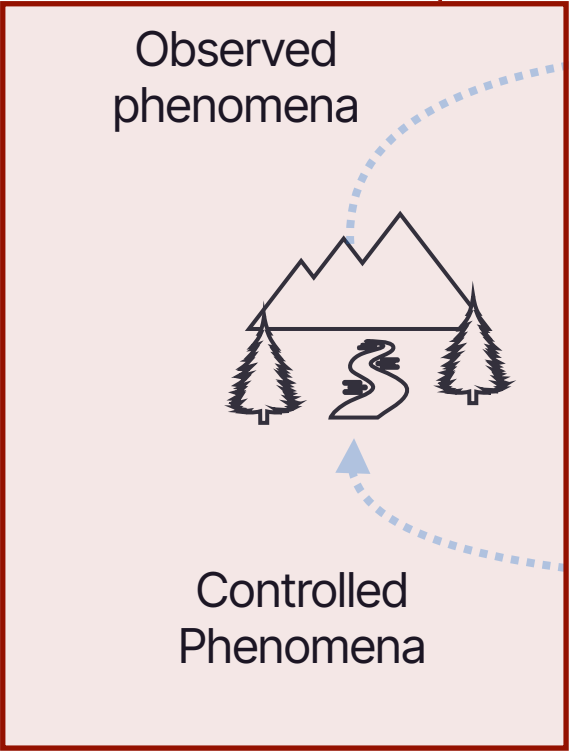


An intelligent system

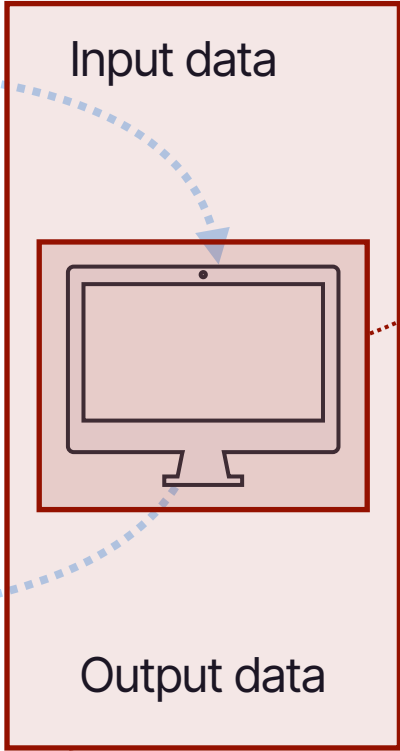
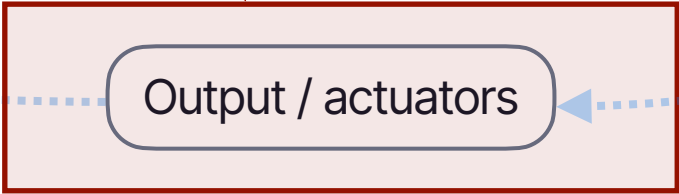
- The intelligent system uses **data to make sense of the world.**
- System **gathers data from outside** and feed the system model.
- System uses **model predictions to decide** which actions to take.

Understanding and acting on the real-world is now system's responsibility!

Requirements



Assumptions



Implementation

Specifications

Case Study: Fall Detection with a Smartwatch



Requirements (REQ)

- Describe how the system should operate.
- Expressed only in terms of the concepts in the world.
- What should happen in real-world, not how software should process it.

Example:

- The watch should call responders when the wearer as fallen.

Specifications (SPEC)

- Describe the expected behavior of a component in terms of input/output.

Examples:

- "Fall detected" warning shall be emitted if model classified it as such.
- If no user response is detected within 30 secs, the system shall call the emergency contact.

Assumptions (ASM)

- Express the relationship of real-world concepts to system's input and output data.

Examples:

- The gyro sensor shall represents person's movement.
- The GPS location on the phone shall represent the user location.
- The smartwatch shall be connected with the phone with bluetooth.

Implementation (IMPL)

- Provide the actual behavior of the software system.
- Should align with what was previously specified (SPEC).



SPEC $\neq$ IMPL $\rightarrow$ Bug

When things go wrong

The Lufthansa 2904 Runway Crash



- 70 people, 64 passengers.
- 2 fatalities.
- 56 injuries.

The Thrust Reverser System

Goal: Automatically decelerate the airplane.

REQ: Thrust reversers shall be activated only in the ground.

ASM.1: The plane is on the ground if at least 6.3T of weight are on each wheel.

ASM.2: The plane is on the ground if wheels are turning faster than 72 knots.

SPEC: Block thrust reversers unless ASM.1 and ASM.2 are satisfied.

In 1993, it was a rainy and windy day...

✗ ASM.1. On wheel was not loaded with proper weight due to the wind.

✗ ASM.2. Wheels were not fast enough due to aquaplaning.

Pilot tried to activate it manually... System overrides.



What went wrong?

<https://ideas.ted.com/justice-in-the-age-of-big-data/>

The program processed historical crime data and calculated, hour by hour, where crimes were most likely to occur.

Idea: police officers would reinforce patrol on more dangerous neighborhoods.

Doesn't focus on the individual, it targets geography, instead.



What went wrong?

<https://www.bbc.com/news/technology-50865437>

REQ? ASM? SPEC?

BBC

Facial recognition fails on race, government study says

20 December 2019

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Facial recognition tools are increasingly being used by police forces

A US government study suggests facial recognition algorithms are far less accurate at identifying African-American and Asian faces compared to Caucasian faces.

Assumption Issues

An intelligent system uses data to interact with the real-world. Thus, we must question assumptions about the data itself.

1. **Unreliable human feedback:** false occurrences of a given event.
User falls, but not an issue.
2. **Data drift:** data changes overtime.
The training data was collected in senior homes, but is increasingly used by young people.
3. **Feedback loops:** system behavior reinforced by data its produce.
Users stop using the watch when gardening due to FPs.
4. **Attacks:** users fake the environment for malicious purposes.
Someone artificially shake the smartwatch while user is sleeping.

Eliciting Requirements

1. Identify Stakeholders

Stakeholder: anyone who have an interest in a project or who may be affected by the project.

- It should include customers and people indirectly affected by the system.
- Conduct a brainstorming session to list all potential stakeholders in the beginning of the project.

What are the stakeholders in the fall detection system? 🤔

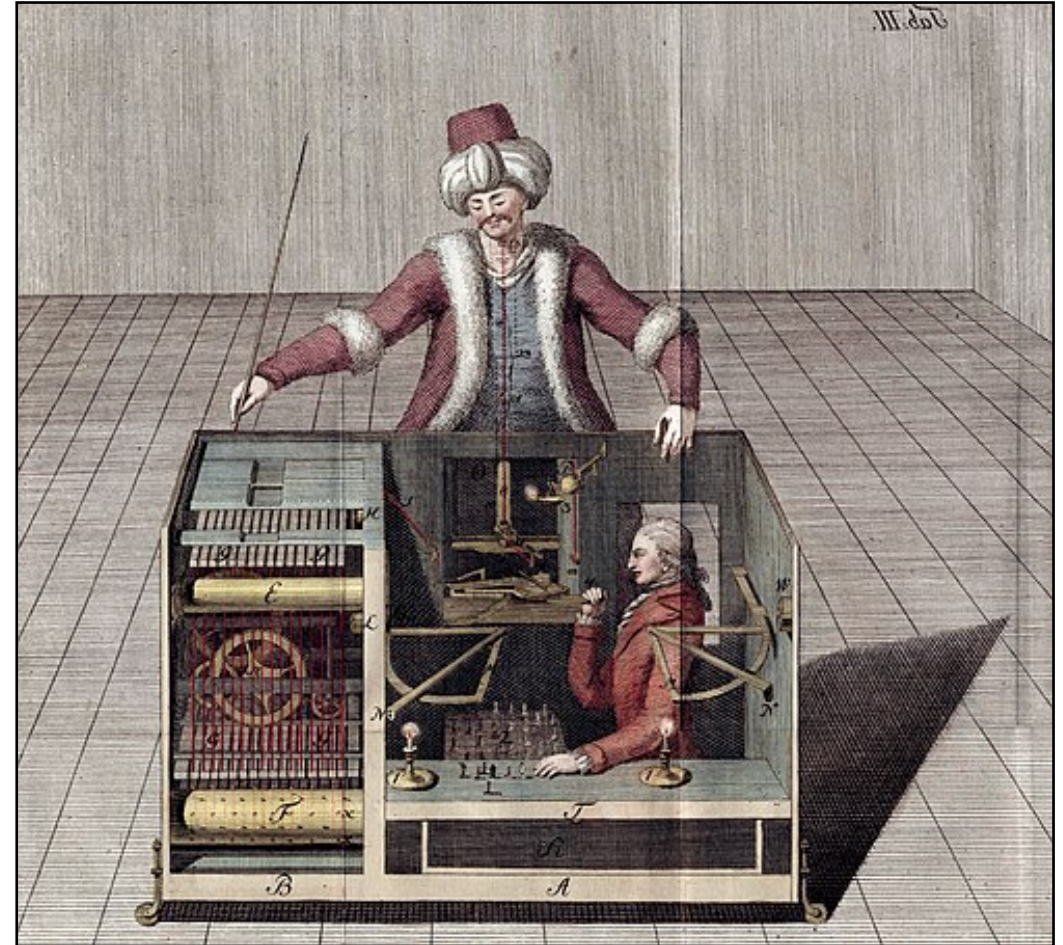
2. Elicit Requirements

1. A **background study** helps understanding the problem and its context.
2. **Interviews** with stakeholders.
 - Ask about their expectations, trade-offs, preferences, concerns, constraints, etc.
 - Supplement it with visuals, storyboards, or prototypes.
3. Explore **additional methods**
 - Surveys, personas, ethnographic studies, checklists, ...

The Wizard of Oz

https://en.wikipedia.org/wiki/Wizard_of_Oz_experiment

A research experiment in which **subjects interact with a computer system** that subjects believe to be autonomous, but **that is actually operated** or partially operated by an unseen **human being**.



3. Negotiate Requirements

Define **which problems to address** and how to prioritize them towards a solution.

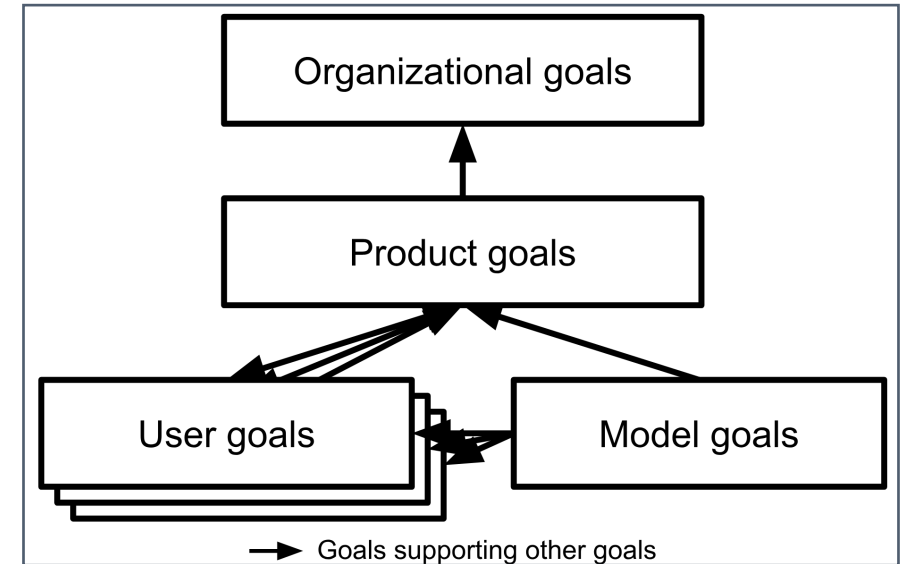
- Group similar concerns, detect conflicting ones, look for alternatives.
- This process can be iterative, and may not satisfy everyone involved.
- Some techniques: [card sorting](#), [affinity diagramming](#), [importance-difficulty matrices](#), etc.
- Overcome with a list of requirements you intend to implement.

Managing Conflicts

Stakeholders $\Rightarrow$ Requirements $\Rightarrow$ Conflicts

- Customers VS 3rd Parties: privacy preferences.
- Product VS Regulators: GDPR.
- Developers VS Operators: software update protocol.

Someone must be in charge to take the decisions.



4. Document Requirements

- Write requirements down to be used during the development process.
- Include assumptions and responsibilities of non-software parts.
 - The system shall prompt the user for confirmation.
 - The user shall wear the smartwatch all the time.
 - The user shall update their emergency contact regularly.
- Prioritize formal documentation for higher-risk projects.

5. Evaluate Requirements

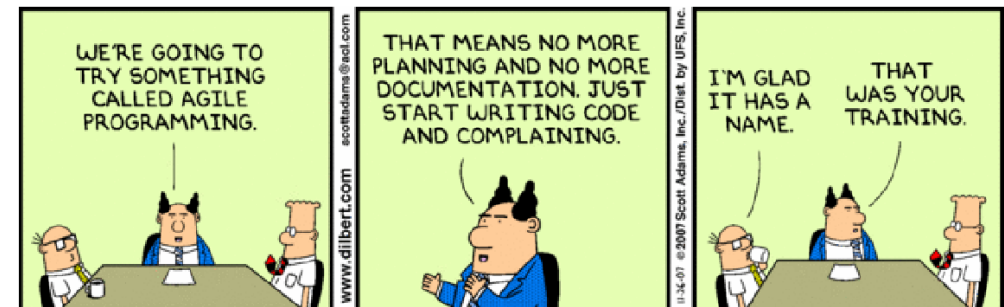
- Should be **analyzed manually**, like code review.
- **Double-check** with stakeholders, ask for inconsistencies.
- **Checklists** to ensure certain qualities are met (privacy, usability, fairness, etc).
- Special attention to ASMs, they should be realistic and complete.
- Look for **AI-specific assumptions**
 - No false positives are acceptable.
 - Data is unbiased and representative.

How much is enough?



Max

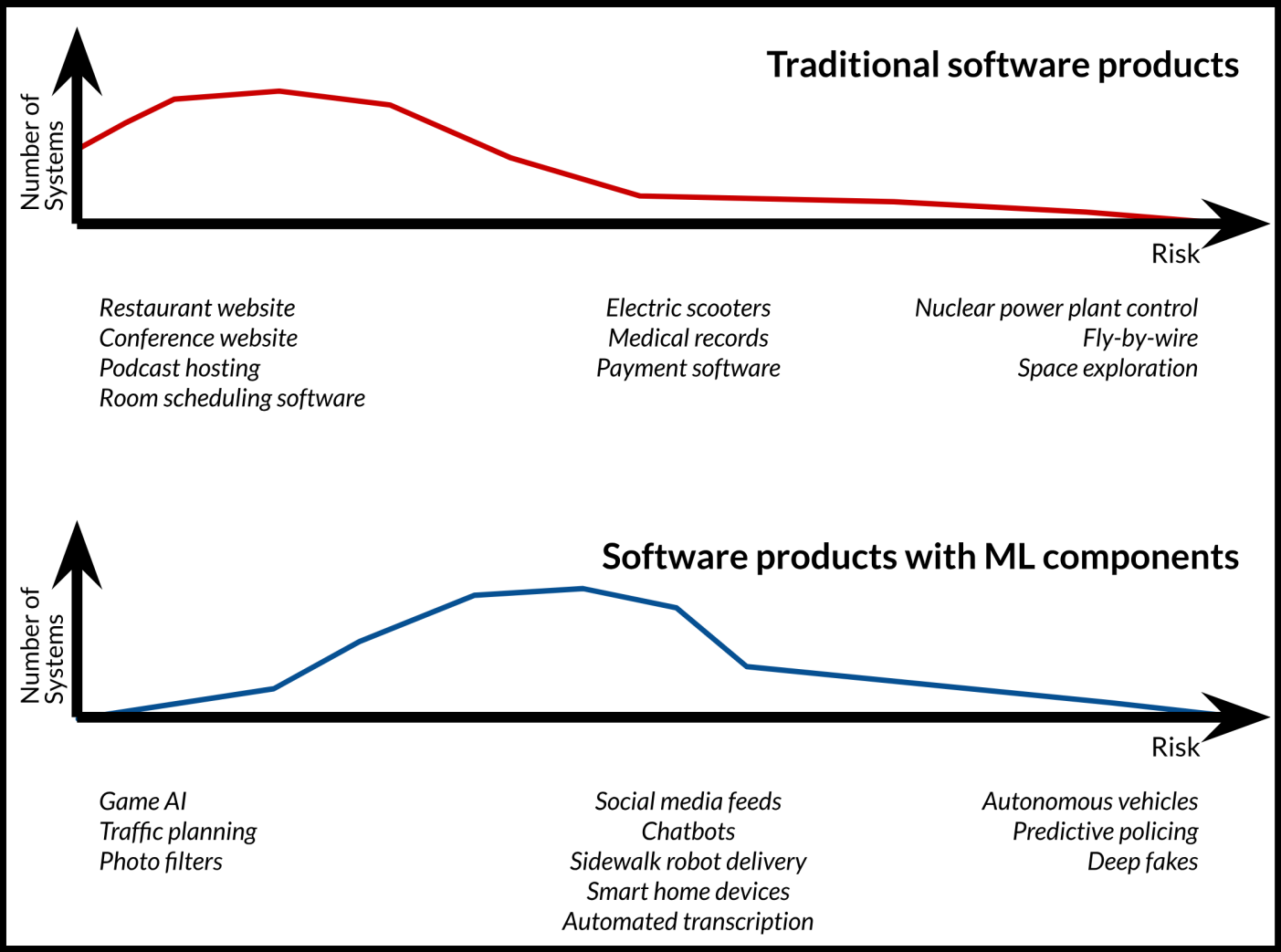
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How much is enough?

- Hard to tell, heavily system dependent.
 - Level of **formalism depends on risk**.
- Intelligent Systems are often ambitious, establishing requirements upfront are very important.
 - REQs, ASMs, and SPECS!
- However, we **can not ignore the experimental nature** of Machine Learning Projects.

One hypothesis



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Thanks!